

Homework 1: August 2026

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Instructions: You are encouraged to discuss and collaborate with your classmates. However, you must explicitly mention at the top of your submission who you collaborated with. Copying is NOT permitted, and solutions must be written independently and in your own words.

General instructions: The exercises below have a theory component and a programming component. Your submission must consist of two files: a single pdf file with all the solutions for the theory component, and a jupyter notebook (ipynb) file with the programming component. The jupyter notebook file must follow the template ipynb file given to you. No other formats will be accepted.

Exercise 1.1. Consider binary classification with $\mathcal{X} = [0, 1]$, and concept class (i.e., assume that the ground truth uses one element from the class)

$$\mathcal{H} = \{h_w(x) = 1_{\{a \leq x \leq b\}} : 0 \leq a \leq b \leq 1\}$$

Theory: For a given dataset

$$\mathcal{S} = \{(x_1, y_1), \dots, (x_n, y_n)\},$$

is the ERM unique? In any case, give a closed form expression for the set of all ERMs for learning the above concept class.

Compute the VC dimension of this hypothesis class.

Programming: For the training/test dataset given to you, implement an ERM and evaluate the accuracy over the test dataset. Also find the precision, recall, and $F1$ score. Note that the test dataset may change at the time of evaluation.

Exercise 1.2. Consider the following set of polynomial classifiers for regression under mean squared error:

$$\mathcal{H}_m = \{h_w(x) = \sum_{i=0}^m a_i x^i : \underline{a} \in \mathbb{R}^{m+1}\}$$

for $1 \leq m \leq 10$.

Theory: Give a closed form expression for the ERM for any fixed m .

Programming: You are given training and test datasets. Split the training set into 80% training and 20% validation set. Using your solution for the theory part, find the ERM for each m , and select the best m using the validation set.

Exercise 1.3. Consider the problem of logistic regression for fashion-MNIST.

Theory: Derive the backprop equations for a perceptron and a fully connected neural network with one hidden layer. Use tanh activation for the hidden layer.

Start with the template code provided to you (using the method discussed in lecture 8). For the fully connected NN with one hidden layer, set the number of hidden neurons to 20. Instead of using the pytorch library for autodiff, use the backprop equations you have derived above and train the model using random shuffling (using the same parameters as in the template code).

Repeat the same with the following optimizers:

- Adagrad ($\beta = 0.99$)
- RMSprop ($\beta = 0.99$)
- Adam ($\alpha = 0.9, \beta = 0.99$)
- Polyak acceleration ($\alpha = 0.95$)
- Nesterov acceleration ($\alpha = 0.95$)

Exercise 1.4. Consider the function

$$f(x_1, x_2, x_3) = \sin(x_1 x_2) + e^{x_1 - x_3} + x_2^2 - x_3$$

- Find the gradient of f at $[1, 1, 1]$ (Theory)
- Use numerical differentiation (using balanced differences), and plot the value of the gradient for $\epsilon \in [10^{-7}, 10^{-1}]$ (Programming)
- Find a computation graph for f . (Theory)
- Using the computation graph above, find the reverse mode autodifferentiation equations and hence compute the gradient. You may directly use expressions for derivatives for sine and exponentials. For each node v , compute v and $\bar{v}$ (Theory)